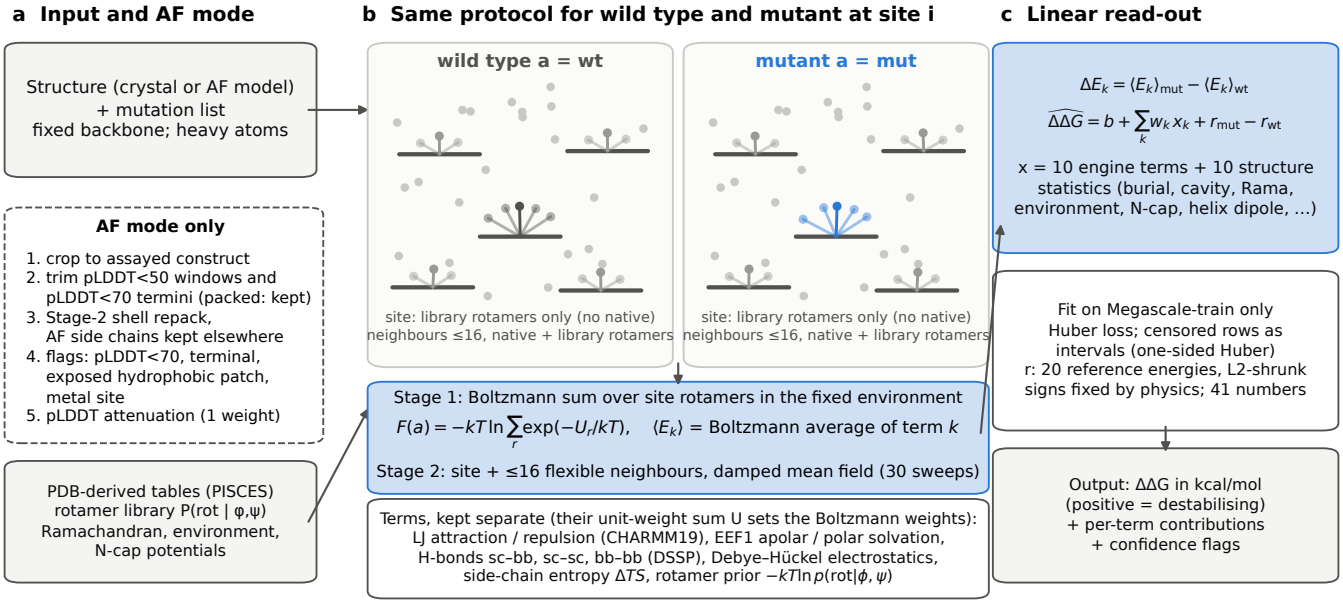

a Input and AF mode
Structure (crystal or AF model)
+ mutation list
fixed backbone; heavy atoms
AF mode only
1. crop to assayed construct
2. trim pLDDT<50 windows and pLDDT<70 termini (packed: kept)
3. Stage-2 shell repack, AF side chains kept elsewhere
4. flags: pLDDT<70, terminal, exposed hydrophobic patch, metal site
5. pLDDT attenuation (1 weight)
PDB-derived tables (PISCES)
rotamer library P(rot | φ,ψ)
Ramachandran, environment,
N-cap potentials
b Same protocol for wild type and mutant at site i
wild type a = wt
mutant a = mut
site: library rotamers only (no native)
neighbours ≤16, native + library rotamers
site: library rotamers only (no native)
neighbours ≤16, native + library rotamers
Stage 1: Boltzmann sum over site rotamers in the fixed environment
$F(a) = -kT \ln \sum_r \exp(-U_r/kT)$, $\langle E_k \rangle$ = Boltzmann average of term k
Stage 2: site + ≤16 flexible neighbours, damped mean field (30 sweeps)
Terms, kept separate (their unit-weight sum U sets the Boltzmann weights):
LJ attraction / repulsion (CHARMM19), EEF1 apolar / polar solvation,
H-bonds sc–bb, sc–sc, bb–bb (DSSP), Debye–Hückel electrostatics,
side-chain entropy ΔTS, rotamer prior $-kT \ln p(\mathrm{rot}|\phi,\psi)$
c Linear read-out
$\Delta E_k = \langle E_k \rangle_{\mathrm{mut}} - \langle E_k \rangle_{\mathrm{wt}}$
$\widehat{\Delta\Delta G} = b + \sum_k w_k x_k + r_{\mathrm{mut}} - r_{\mathrm{wt}}$
x = 10 engine terms + 10 structure statistics (burial, cavity, Rama, environment, N-cap, helix dipole, ...)
Fit on Megascale-train only
Huber loss; censored rows as intervals (one-sided Huber)
r: 20 reference energies, L2-shrunk
signs fixed by physics; 41 numbers
Output: ΔΔG in kcal/mol
(positive = destabilising)
+ per-term contributions
+ confidence flags